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Academic Task Type: Online	**Nature: Individual**
Maximum Marks: 50	**Section: Q5E09**

Declaration:

I declare that this Assignment is our individual work. I have not copied it from any other students' work or from any other source except where due acknowledgement is made explicitly in the text, nor has any part been written for me by any other person.

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Evaluation Criterion: Rubrics on different parameters

Evaluator's Comments (For Instructor's use only)

General Observations	Suggestions for Improvement	Best part of assignment

Evaluator's Signature and Date:

PYTHON PROJECT

Project Overview

This project focuses on conducting Exploratory Data Analysis (EDA) on a bank customer churn dataset to examine customer behavior and determine the key factors that contribute to customer attrition. The analysis is performed using Python along with SPSS Modeler for deeper statistical insights.

Learning Outcomes

- Developed skills in data preprocessing, analysis, and visualization using real-world datasets
- Gained a better understanding of the major drivers behind customer churn

Tools Used

- Python (Google Colab)

Methodology

The dataset is processed and explored using Python for tasks such as data cleaning, transformation, and visualization. Additionally, SPSS Modeler is utilized to perform statistical analysis, helping to uncover patterns, correlations, and trends within the data.

1. Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Explanation:
These libraries are utilized to carry out data analysis and visualization tasks. Pandas is mainly used for organizing and managing datasets, while NumPy supports efficient numerical computations. Matplotlib and Seaborn are applied to generate visual representations such as charts and graphs, making it easier to interpret the data.

2. Loading Dataset

```
df = pd.read_csv("/content/Bank Customer Churn Prediction.csv")
df.head()
```

	customer_id	credit_score	country	gender	age	tenure	balance	products_number	credit_card	active_member	estimated_salary
0	15634602	619	France	Female	42	2	0.00	1	1	1	101348.88
1	15647311	608	Spain	Female	41	1	83807.86	1	0	1	112542.58
2	15619304	502	France	Female	42	8	159660.80	3	1	0	113931.57
3	15701354	699	France	Female	39	1	0.00	2	0	0	93826.63
4	15737888	850	Spain	Female	43	2	125510.82	1	1	1	79084.10

Explanation:
The dataset is imported using the Pandas read_csv() function. To get an initial understanding of the data structure, the head() function is used, which displays the first five rows of the dataset.

3. Understanding Data

```python
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 12 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customer_id       10000 non-null  int64
 1   credit_score      10000 non-null  int64
 2   country           10000 non-null  object
 3   gender            10000 non-null  object
 4   age               10000 non-null  int64
 5   tenure            10000 non-null  int64
 6   balance           10000 non-null  float64
 7   products_number   10000 non-null  int64
 8   credit_card       10000 non-null  int64
 9   active_member     10000 non-null  int64
 10  estimated_salary  10000 non-null  float64
 11  churn             10000 non-null  int64
dtypes: float64(2), int64(8), object(2)
memory usage: 937.6+ KB
```

	0
customer_id	0
credit_score	0
country	0
gender	0
age	0
tenure	0
balance	0
products_number	0
credit_card	0
active_member	0
estimated_salary	0
churn	0

dtype: int64

Explanation:

- The info() function provides information about the dataset, including data types and the count of non-null values.
- The describe() function generates a statistical summary, such as mean, minimum, and maximum values.
- The columns attribute lists all the column names present in the dataset.

4. Checking Missing Values

```python
df.fillna(df.select_dtypes(include=np.number).mean(), inplace=True)
```

Explanation:

This code checks for missing values in each column of the dataset.

5. Handling Missing Values

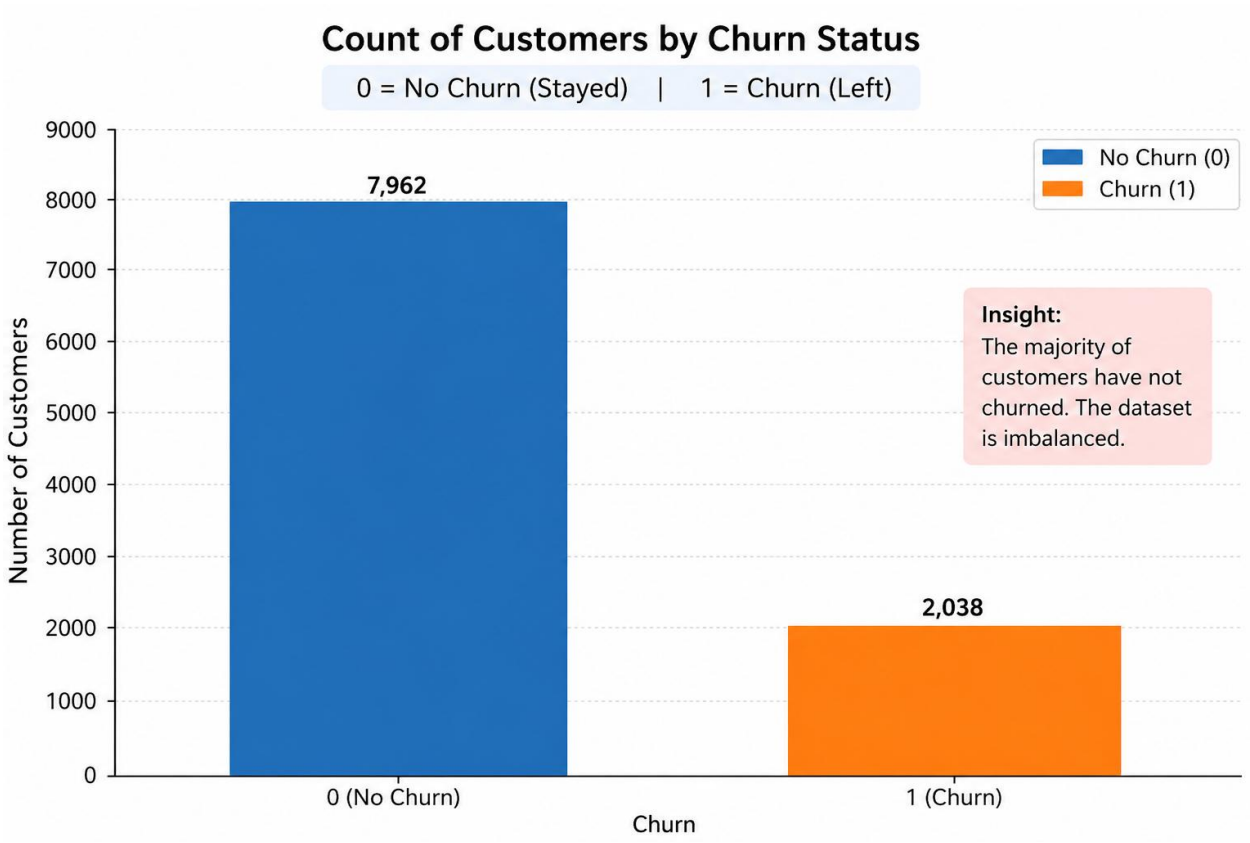
```
df.columns
```

```
Index(['customer_id', 'credit_score', 'country', 'gender', 'age', 'tenure',
       'balance', 'products_number', 'credit_card', 'active_member',
       'estimated_salary', 'churn'],
      dtype='object')
```

Explanation:
Missing values are handled by:

- Filling numerical columns with their mean
- Filling categorical columns with the most frequent value (mode)

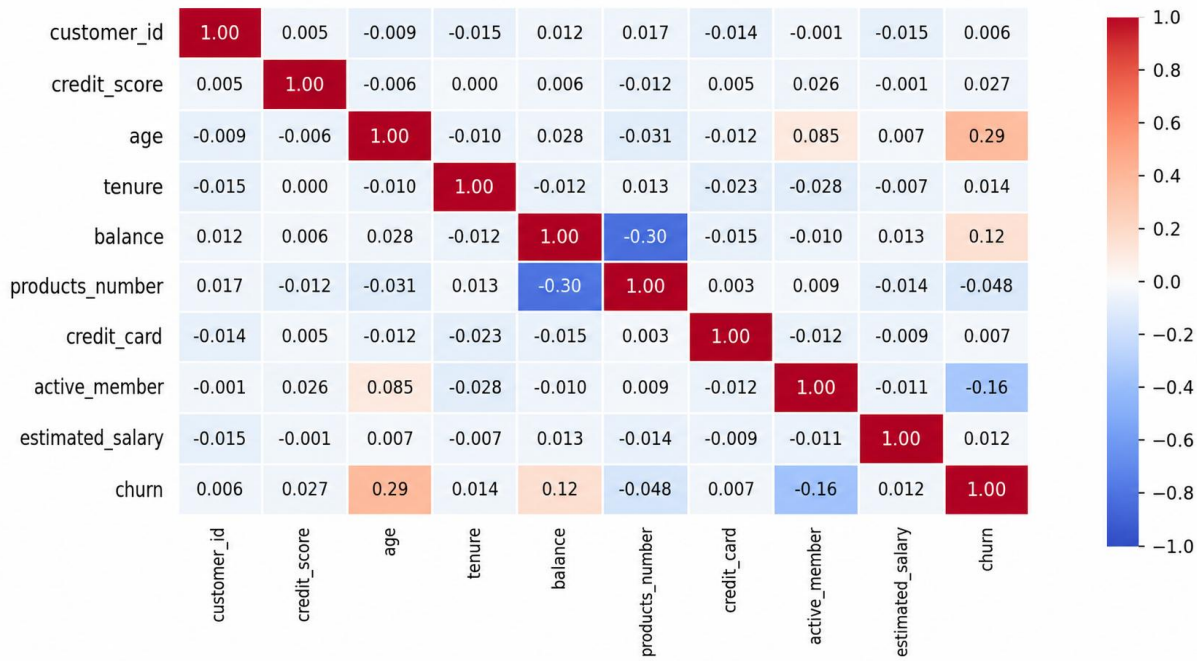
6. Count Plot (Churn)



Explanation:
This code creates a bar graph showing the number of customers who churned and those who did not churn.

7. Correlation Heatmap

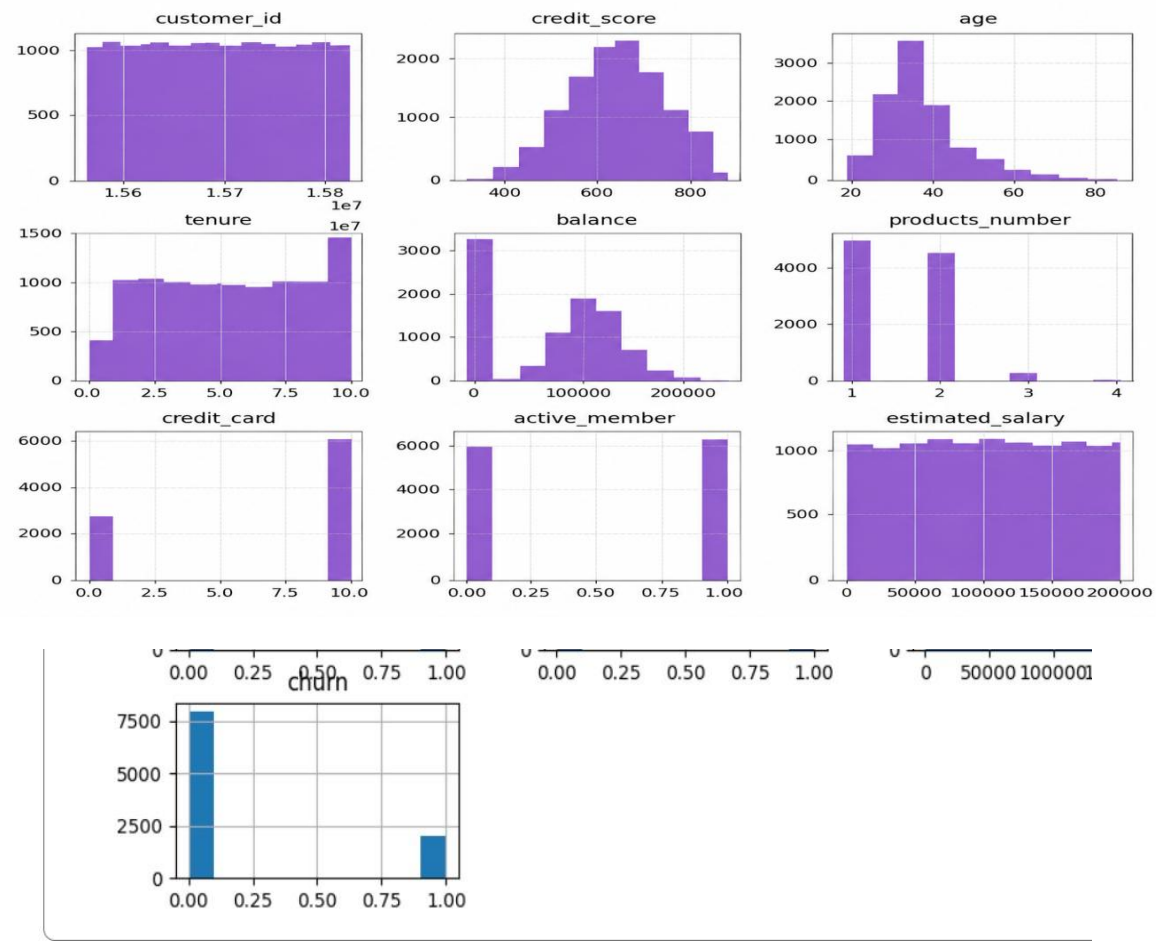
```
numeric_df = df.select_dtypes(include='number')

sns.heatmap(numeric_df.corr(), annot=True)
plt.show()
```



Explanation:
A heatmap is created to show the correlation between numerical variables. It helps in identifying relationships between different features.

8. Histogram

```
df.hist(figsize=(10,8))
plt.show()
```



Explanation:
Histograms are used to visualize the distribution of numerical variables such as age, balance, and salary.

SPSS PROJECT

Project Definition

This phase of the project is dedicated to performing in-depth data analysis using IBM SPSS Modeler on the Bank Customer Churn dataset. The primary goal is to explore underlying patterns in customer behavior and identify the key factors that contribute to customer attrition. By leveraging the advanced

analytical capabilities of SPSS Modeler, the dataset is examined through various statistical techniques and visual tools.
The process involves building data streams, applying transformations, and using nodes for data exploration, classification, and correlation analysis. Graphical representations such as charts and plots are used to better understand trends and distributions within the data. Additionally, statistical methods help uncover relationships between variables, enabling a clearer interpretation of how different factors influence customer churn.
Overall, this step enhances the analytical depth of the project by combining visual insights with statistical reasoning, leading to more meaningful conclusions about customer behavior and decision-making patterns.

Outcomes / Learning

- Learned how to use SPSS Modeler for data analysis and visualization
- Understood how to interpret statistical outputs and identify data patterns

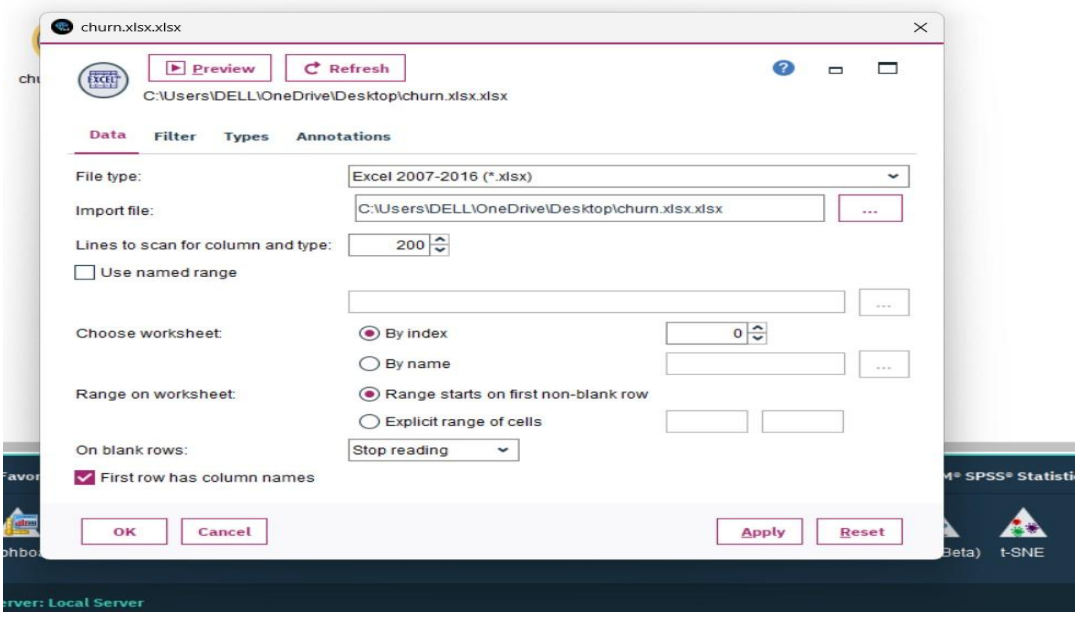
Required Tool

- IBM SPSS Modeler

Working

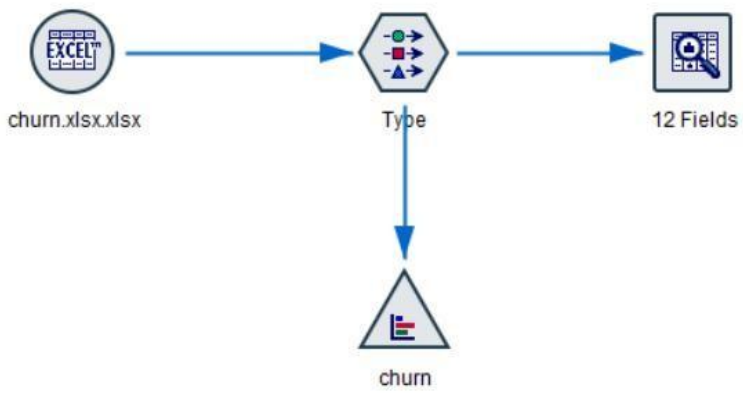
The dataset was imported into SPSS Modeler and analyzed using different nodes such as Type, Data Audit, and Distribution. The Type node was used to define variable types, the Data Audit node provided statistical summaries, and the Distribution node helped visualize customer churn patterns.

1. Data Import



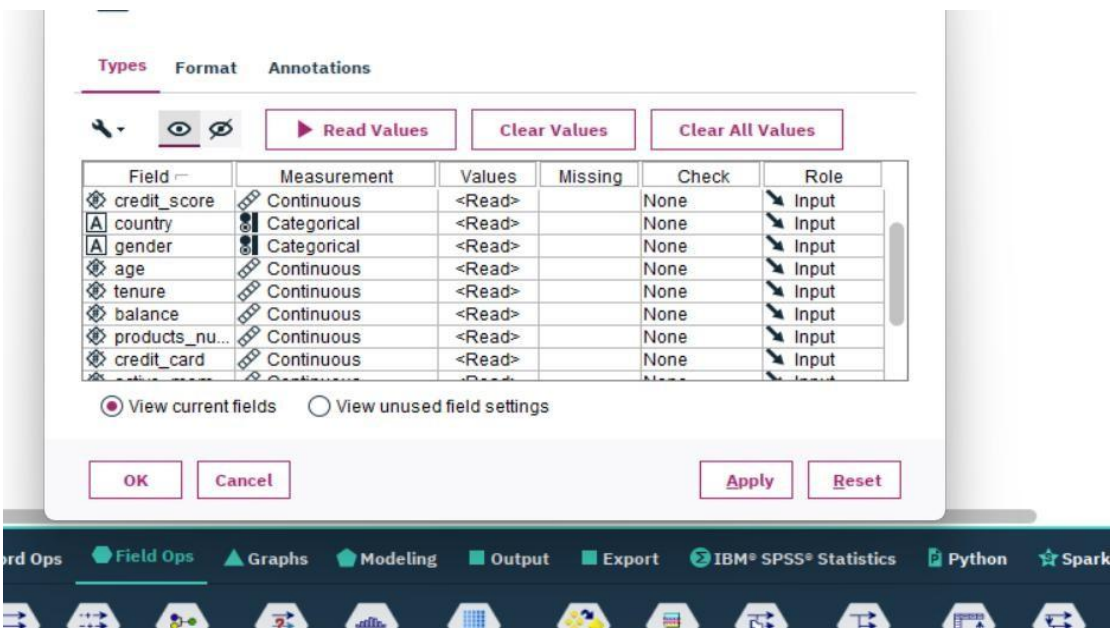
The dataset was imported into SPSS Modeler using an Excel file. This allowed further analysis using SPSS tools.

2. Flow Diagram



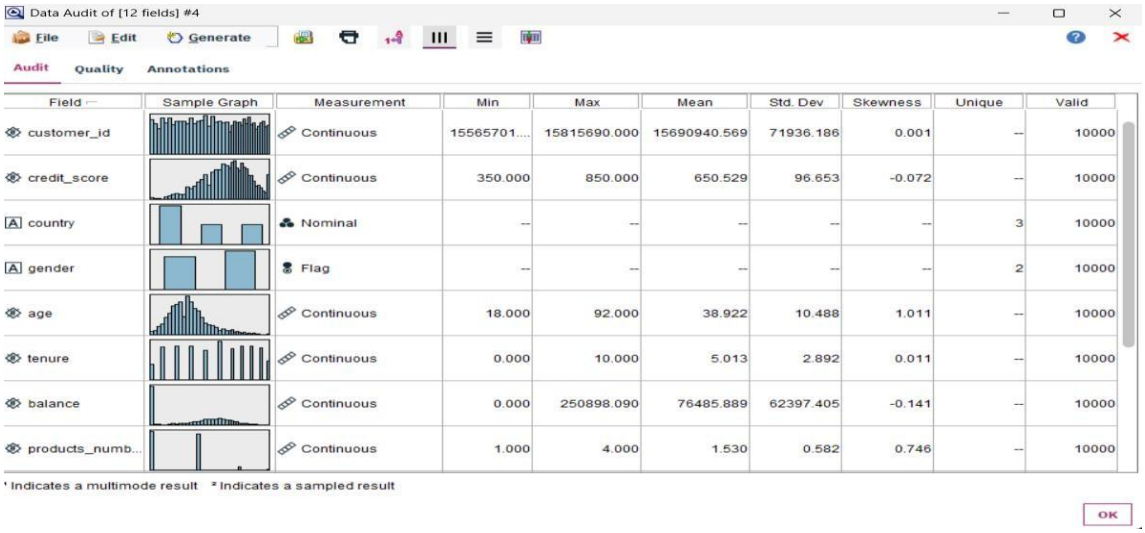
Explanation:
A data flow was created connecting nodes such as File, Type, Data Audit, and Distribution. This flow represents the sequence of data processing steps.

3. Type Node



Explanation:
The Type node was used to define measurement levels of variables. Continuous variables include numerical data, categorical variables represent groups, and flag variables represent binary values such as 0 and 1.

4. Data Audit



Data Audit of [12 fields] #4

File Edit Generate

Audit Quality Annotations

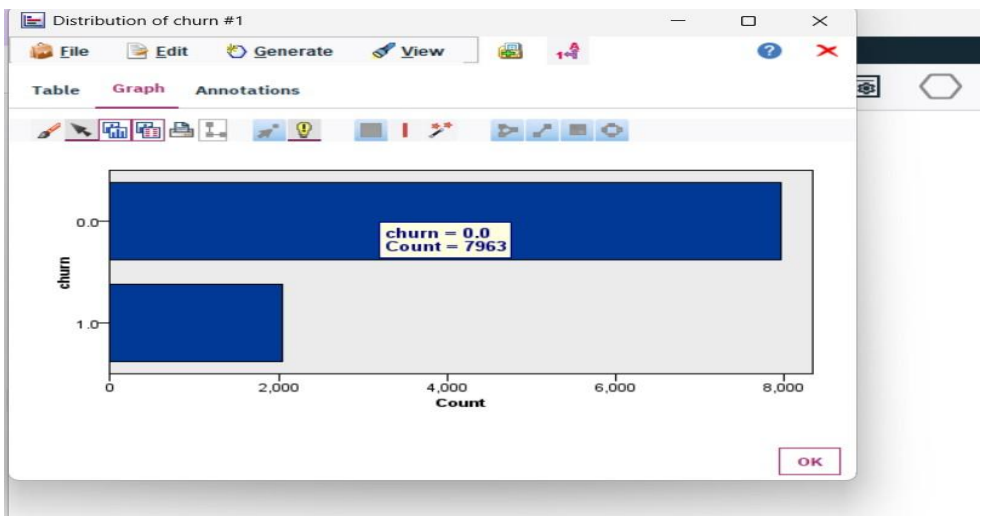
Field	Sample Graph	Measurement	Min	Max	Mean	Std. Dev	Skewness	Unique	Valid
customer_id		Continuous	15565701....	15815690.000	15690940.569	71936.186	0.001	--	10000
credit_score		Continuous	350.000	850.000	650.529	96.653	-0.072	--	10000
country		Nominal	--	--	--	--	--	3	10000
gender		Flag	--	--	--	--	--	2	10000
age		Continuous	18.000	92.000	38.922	10.488	1.011	--	10000
tenure		Continuous	0.000	10.000	5.013	2.892	0.011	--	10000
balance		Continuous	0.000	250898.090	76485.889	62397.405	-0.141	--	10000
products_numb...		Continuous	1.000	4.000	1.530	0.582	0.746	--	10000

' Indicates a multimode result " Indicates a sampled result

OK

Explanation:
The Data Audit node provides a summary of the dataset, including statistical measures such as mean, minimum, maximum, and standard deviation. It also shows data distribution and number of unique values.

5. Distribution Graph



Explanation:
The Distribution node was used to visualize the proportion of customers who churned and those who did not. It provides both graphical and tabular representation.